Haylie Wu

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Education

New England Conservatory of Music - *M.S. in Flute Performance and Music Technology* **GPA: 4.0** Aug 2025 - May 2027
Northwestern University - ***B.S. in Computer Science*** *and B.M. in Flute Performance* **GPA: 3.98** Sep 2021 - Jun 2025

Professional Experience

MathWorks Jun 2023 - Sep 2023
Software Engineer Intern
Tech stack: JavaScript, MATLAB, C++, Node.js, Express, MongoDB

- Designed and built a 2D parser-combinator architecture in Simulink that composes reusable parsers to validate tabular Excel test data against structural and domain rules, with multi-error handling that reports every invalid cell via inline highlighting and diagnostics.
- Built a centralized asynchronous initialization layer for a MATLAB test framework, replacing scattered sequential setup with concurrent loading and shared dependency handling to cut startup latency by ~30% while improving maintainability and testability.
- Shipped two full-stack features for MATLAB R2024a: persistent testing-session configurations and keyword filtering, working across a large legacy codebase and reaching 90% test coverage.
- Implemented a Node/Express/MongoDB backend for an internal bug-triage tool used in sprint planning, adding a configurable scoring system to rank and prioritize incoming bugs.

Northwestern University - Center for Connected Learning (NetLogo Web) Sep 2022 - Jun 2024
Software Engineer & Open Source Contributor
Tech stack: TypeScript, Svelte, CoffeeScript, Ractive.js, HTML, CSS, Jest

- Built and shipped a TypeScript NetLogo Web editor with CodeMirror 6, replacing the legacy CodeMirror 5 implementation with a new UI, editor interactions, syntax highlighting, and autocompletion; wrote Jest tests simulating cursor, keyboard, and editing behavior.
- Developed an interactive drawing tool for NetLogo Web that lets users draw, edit, save, and reuse custom agent shapes, paired with a shape library for managing simulation icons across models.
- Contributed to ChatLogo, a GPT-4-powered in-editor assistant for NetLogo syntax and platform questions.
- Automated testing, version management, npm package publishing, and deployment with GitHub Actions workflows.

SKUY May 2022 - Aug 2024
Lead Backend Software Engineer | Northwestern Garage-backed startup, ~15 engineers
Tech stack: React Native, Flask, PostgreSQL, Python, JavaScript, Heroku, Firebase, Puppeteer, Selenium

- Built and maintained the backend and several frontend features for a mobile social platform, developing Flask/PostgreSQL APIs and scheduled web-scraping pipelines that powered a dynamic news feed for 1,000+ registered users.
- Improved app performance by redesigning content APIs around pagination and lazy loading, replacing large single-shot responses for feeds and profiles, cutting initial load time by ~35%.
- Migrated application data from PostgreSQL to Firebase and built a transition path that let existing beta users carry over without re-registering.
- Owned deployment and onboarding as the team's lead engineer, managing Heroku/App Store releases and writing the API and workflow docs new engineers ramped up on.

Northwestern University - Algorithm Design Research Jun 2022 - Sep 2022
Research Intern, advised by Prof. Samir Khuller

- Developed and analyzed approximation algorithms for active-time scheduling on heterogeneous machines using dynamic programming and lazy-activation techniques, and ran experiments in **Python** and **Gurobi** to measure how close the approximations came to the optimum.

Projects

Cookie Flute Studio | Next.js, TypeScript, Web Audio API, MusicXML, AWS Jun 2026 - Present

- Built a real-time score-following engine that aligns autocorrelation-based pitch detection from live audio with parsed MusicXML scores to provide per-note intonation feedback as students play.
- Developed a full-stack practice platform with repertoire management, score annotation, practice tracking, and integrated training tools, backed by a serverless AWS pipeline (S3, Lambda, API Gateway, DynamoDB) for storing and analyzing practice recordings.
- Designing a practice-insights engine that aggregates note-level pitch data across sessions to identify recurring problem passages and surface targeted practice recommendations.

Skills

Languages: Python, TypeScript/JavaScript, Java, C++, SQL, MATLAB
Frameworks & Cloud: React/Next.js, React Native, Node.js/Express, Flask, PostgreSQL, MongoDB, Firebase, AWS, Git